

How to Explain DRC Validation

1. What is DRC?

DRC = Design Rule Check. It verifies that the built chip geometry follows the layout rules before we compare topologies.



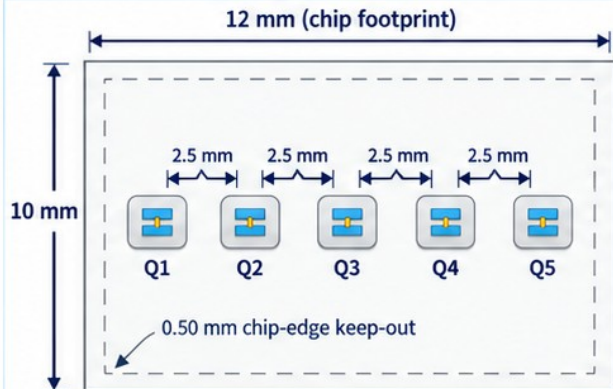
2. Rules Checked

Rule	Value
Chip footprint	12 mm × 10 mm
Baseline pitch	2.5 mm center-to-center
Nominal CPW geometry	width = 10 μ m, gap = 6 μ m
Minimum accepted CPW rule	width $\geq$ 8 μ m, gap $\geq$ 4 μ m
Minimum qubit pocket gap	$\geq$ 0.30 mm
Chip-edge keep-out	$\geq$ 0.50 mm

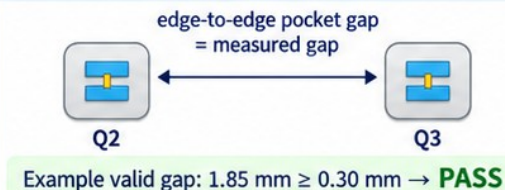
Pitch is center-to-center spacing; DRC gap is edge-to-edge geometry spacing.

3. How PASS is determined

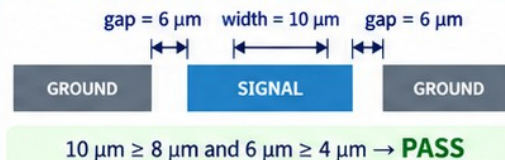
A. Chip layout (top view)



B. Pocket gap check (zoom-in)

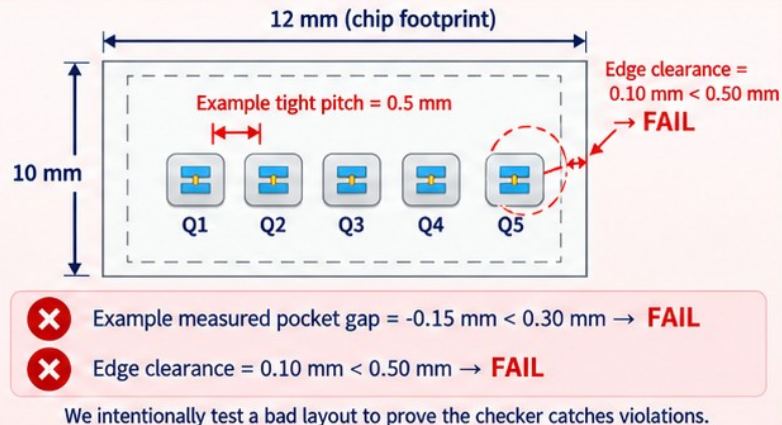


C. CPW cross-section check (top view)



4. How FAIL is detected

Invalid layout example (top view)



What to say

1. We first build the actual Metal geometry, not just idealized points.
2. Then DRC checks spacing, CPW width/gap, and chip-edge keep-out against one common rule set.
3. Our normal baseline passes, while an intentionally tight layout fails, which proves the checker works.

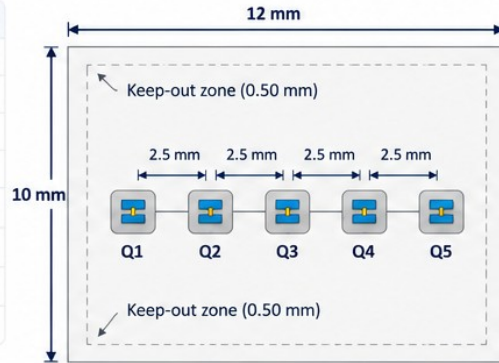
Task 02 — Coupling Map / Topology Exploration

Two physical topologies, one rule set, different workloads

Same 5 qubits
Same design rules
Different connectivity
Different use cases

1 Common Design Rules (used for both topologies)

Chip footprint	12 mm × 10 mm
Qubit type	TransmonPocket
Baseline pitch / radius	2.5 mm
CPW width (nominal)	10 μ m
CPW gap (nominal)	6 μ m
Minimum CPW width	$\geq 8 \mu$ m
Minimum CPW gap	$\geq 4 \mu$ m
Minimum qubit pocket gap	≥ 0.30 mm
Chip-edge keep-out	≥ 0.50 mm



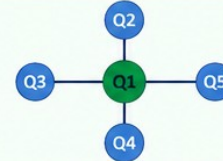
2 Topology Candidates

Linear Chain Topology



- **Coupling map:** nearest-neighbor
- **Edges:** (Q1–Q2), (Q2–Q3), (Q3–Q4), (Q4–Q5)
- **Target workload:** local interactions, Trotter/QAOA-style circuits
- **Advantage:** simple routing, low risk
- **Limitation:** long-range interactions require multi-hop paths

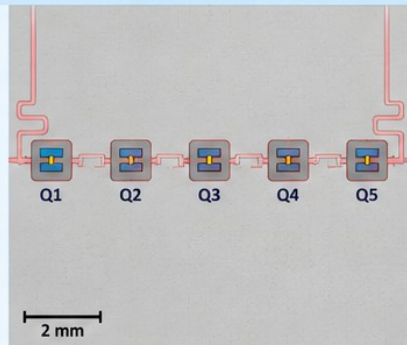
Star / Hub Topology



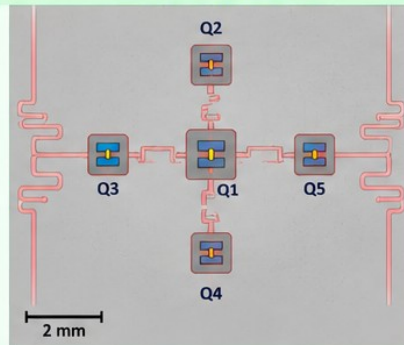
- **Coupling map:** central qubit to all others
- **Edges:** (Q1–Q2), (Q1–Q3), (Q1–Q4), (Q1–Q5)
- **Target workload:** GHZ-state preparation, hub-and-spoke interactions
- **Advantage:** direct connectivity to all qubits
- **Limitation:** higher-routing density near center, potential crosstalk risk

3 Physical Layouts (Qiskit Metal Renders)

Linear Chain Layout



Star Topology Layout



4 Topology Graph Metrics

Metric	Linear Chain	Star / Hub
Number of qubits	5	5
Number of edges	4	4
Maximum degree	2	4
Graph diameter	4	2
Average shortest path	2.0	1.6
Typical routing length (mm)*	~ 8.5	~ 12.3
Expected crossings	0	0 (with diagonal placement)

*Example values from 2.5 mm pitch/radius (exact values from metal_analysis).

5 Workload Fit (Graph Distance Heatmap)

	Linear Chain						Star / Hub				
	Q1	Q2	Q3	Q4	Q5		Q1	Q2	Q3	Q4	Q5
Q1	0	1	2	3	4		0	1	1	1	1
Q2	1	0	1	2	3		1	0	2	2	2
Q3	2	1	0	1	2		1	2	0	2	2
Q4	3	2	1	0	1		1	2	2	0	2
Q5	4	3	2	1	0		1	2	2	2	0

6 Key Takeaways (Task 02)

- ✓ Implemented two physical topologies (Linear and Star) using the same 5 qubits and the same design rules.
- ✓ Motivated each topology with a target workload (local/Trotter vs. GHZ/hub).
- ✓ Verified that both layouts are routable and rule-compliant (detailed DRC in later tasks).

Topology choice is a workload-versus-hardware trade-off: better logical connectivity can increase routing and EM complexity.

Task 3 – Physical Layout Optimization

Optimize qubit placement for each topology by sweeping a layout parameter and minimizing a cost function (total coupler/resonator length) while satisfying design rules (DRC).

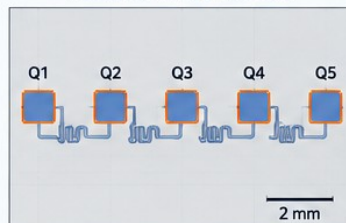
- Task 1: Qubit Design
- Task 2: Topology Implementation
- Task 3: Physical Layout Optimization**
- Task 4: ...

1 What We Optimize

- Topologies:**
 - Linear chain (Q1 – Q2 – Q3 – Q4 – Q5)
 - Star (Q1 at center connected to Q2, Q3, Q4, Q5)
- Layout parameter (swept):**
 - Linear: neighbor qubit pitch (distance between adjacent qubits)
 - Star: hub-to-leaf radius (distance from center to outer qubits)
- Cost function:**
 - Minimize total_route_length_mm subject to drc_violations == 0 (feasible layouts only)
- Sweep range:**
 - 0.9 mm – 2.8 mm (step 0.1 mm)

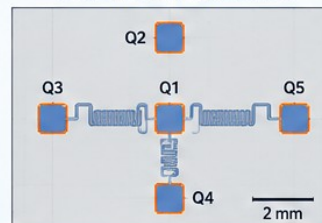
2 Generated Layouts (Qiskit Metal)

Linear Chain (optimized)



- 5 transmon qubits in a line
- 4 CPW couplers (Q1–Q2–Q3–Q4–Q5)
- Qubit pitch (swept): 0.9 – 2.8 mm
- Best pitch: 1.2 mm

Star Topology (optimized)

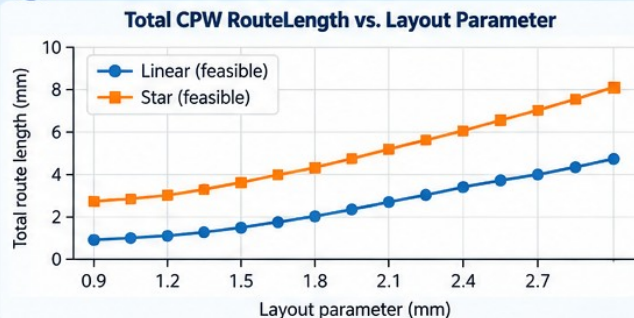


- Q1 at center, connected to Q2, Q3, Q4, Q5
- 4 CPW couplers
- Hub-to-leaf radius (swept): 0.9 – 2.8 mm
- Best radius: 1.5 mm

3 Design Rules (DRC)

Rule	Value
Chip size	12 mm × 10 mm
CPW center width	10 μm (0.010 mm)
CPW gap	6 μm (0.006 mm)
Minimum qubit gap	0.30 mm
Edge keepout	0.50 mm
Route clearance	0.10 mm
No route crossings	Yes
All routes connected	Yes

4 Parameter Sweep Results



5 Optimization Results

Topology	Optimal parameter (mm)	Total route length (mm)	DRC status
Linear	1.2	1.400	PASS
Star	1.5	3.356	PASS

Result: The linear topology achieves a shorter total coupler length at its optimal spacing, while the star topology requires a longer total length due to the central connections.

6 Full Sweep Data

Columns in metal_sweep.csv	Description
topology	linear or star
pitch_mm	sweep value (for report compatibility)
layout_parameter_mm	swept spacing/radius
sweep_variable	name of layout parameter
drc_violations	number of DRC violations
drc_notes	details if any
total_route_length_mm	total CPW length (cost metric)
longest_route_mm	maximum single route length
crossings	number of route crossings
min_route_clearance_mm	minimum spacing between routes
is_feasible	True if drc_violations == 0
optimization_cost_mm	cost used for optimization (inf if not feasible)
cost_function	minimize total_route_length_mm subject to drc_violations == 0

Key Takeaways

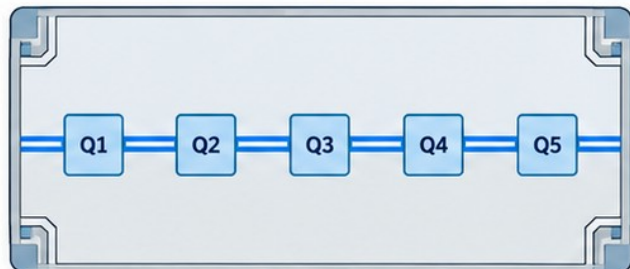
- ✓ We performed a parameter sweep over qubit spacing for two topologies.
- ✓ We built real Qiskit Metal layouts and ran full geometric DRC checks.
- ✓ We minimized total CPW route length among feasible layouts.
- ✓ Linear topology gives a more compact layout (1.400 mm) than star (3.356 mm) under the same design rules.
- ✓ All results, figures, and CSV files are saved in the repository for reproducibility.

Task 4 — EM-Aware Routing Considerations

Routing CPW couplers for Linear and Star 5-qubit topologies with crossing, clearance, crosstalk, and resonator-length considerations

Linear Topology

5-qubit chain with nearest-neighbor CPW couplers



CPW route (signal) Qubit (transmon)

1 mm

Routing Metrics (Linear)

Total route length:	1.400 mm
Longest route:	0.350 mm
Crossings:	0
Airbridges needed:	0
Minimum route clearance:	0.828 mm
Crosstalk-risk proxy:	Lower

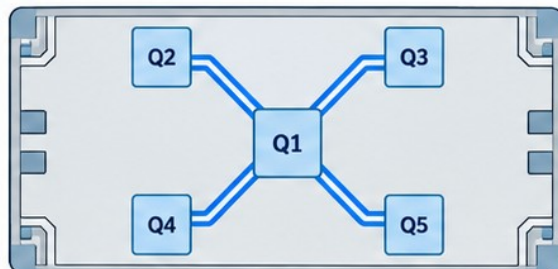
✓ Distributed local routing, lower congestion



Both selected layouts: 0 crossings → no crossing airbridges required

Star Topology

5-qubit star with central hub (Q1)



CPW route (signal) Qubit (transmon)

1 mm

Routing Metrics (Star)

Total route length:	3.356 mm
Longest route:	0.839 mm
Crossings:	0
Airbridges needed:	0
Minimum route clearance:	0.368 mm
Crosstalk-risk proxy:	Moderate / watch

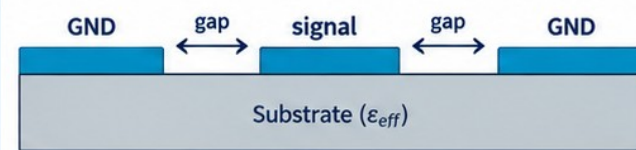
⚠ Hub concentration raises congestion and crosstalk risk



Linear has larger route clearance → lower geometric crosstalk risk

EM Interpretation

CPW geometry, resonator-length considerations, and EM notes



CPW cross-section (not to scale) ← gap → = clearance

λ/4 Resonator Length (approx.)

$$f_{\lambda/4} \approx \frac{c}{4L \sqrt{\epsilon_{eff}}}$$

$\epsilon_{eff} \approx 6$

Approx. 6 GHz $\lambda/4$ length ≈ 5.10 mm

Approx. 6 GHz $\lambda/2$ length ≈ 10.20 mm

Our routed elements are short CPW couplers, not tuned $\lambda/4$ resonators.

Full-wave EM extraction is future work.



Parallel or closely spaced routes can increase unintended coupling.



Star offers hub connectivity but concentrates routing near Q1

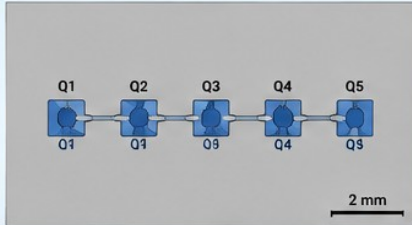


Task 4 takeaway: Both layouts are routeable, but Linear is cleaner physically, while Star trades simplicity for centralized connectivity.

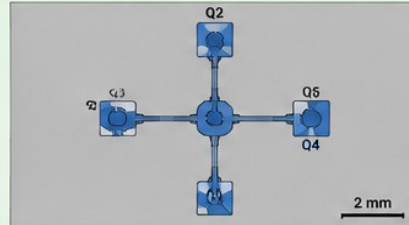
Task 5 — DRC & Manufacturability Validation

Verify design rule compliance and document the iteration process

Linear Topology (Final – 1.2 mm pitch)



Star Topology (Final – 1.5 mm radius)



Design Rule Check (DRC) Summary

Rule	Requirement	Final Linear	Final Star
Chip size	12 mm × 10 mm	✓	✓
Qubit pocket gap	≥ 0.30 mm	✓	✓
CPW width / gap (nominal)	10 μm / 6 μm	✓	✓
Minimum CPW width / gap	≥ 8 μm / 4 μm	✓	✓
Edge keepout	≥ 0.50 mm	✓	✓
Route clearance	≥ 0.15 mm	✓	✓
Route crossings	0 (target)	✓	✓
Endpoint / obstruction checks	Valid	✓	✓

FINAL LINEAR
0 DRC violations
✓ **PASS**

FINAL STAR
0 DRC violations
✓ **PASS**

Design Iteration Log (Selected Examples)

1. Too tight (Linear 0.9 mm)



- Build failure
- Qubits too close / routing issues

2. Qubits crowded (Star 0.9 / 1.2 mm)



- Pocket-gap violation
- DRC failure

3. Too wide (Linear 2.8 mm)



- 0.50 mm edge-keepout violation
- Exceeds chip boundary

FAIL

→ Diagnose

→ Change geometry

→ Rerun DRC

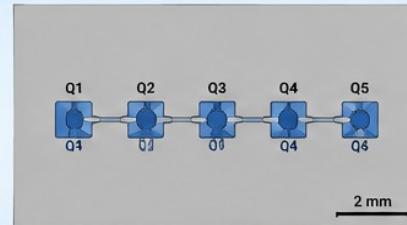
→ PASS

Failed iterations helped us define the feasible design region and guided the final optimized layouts.

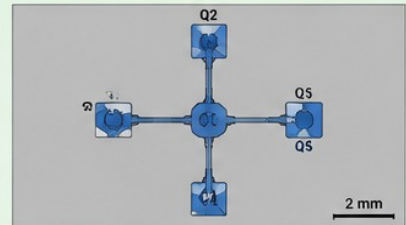
Task 6 — Final Comparison & Recommendation

Compare the optimized designs and recommend the best candidate

Linear Topology (1.2 mm pitch)



Star Topology (1.5 mm radius)



Quantitative Comparison

Metric	Linear	Star	Comments
Selected geometry	1.2 mm pitch	1.5 mm radius	From parameter sweep
Total routing length (mm)	1.400	3.356	Shorter is better
Longest route (mm)	0.350	0.839	Shorter is better
Route crossings	0	0	No crossing airbridges needed
Airbridges needed	0	0	Not required in final designs
Minimum route clearance (mm)	0.828	0.368	Larger is better
DRC violations	0 (PASS)	0 (PASS)	Both compliant
Graph max degree	2	4	Star enables hub connectivity
Typical workload fit	Local / Trotter	GHZ / hub	Different application strengths

Key Takeaways

Why Linear is the better fabrication-development candidate

- ✓ Substantially less total routing
- ✓ Shorter longest-route proxy
- ✓ >2× larger route-clearance margin
- ✓ Zero crossings / zero airbridges
- ✓ Lower routing congestion
- ✓ Simpler first fabrication-development candidate

When Star is still preferred

- ✓ Direct hub-to-leaf connectivity
- ✓ Useful for GHZ-style experiments
- ✓ Central control operations
- ✓ Higher graph connectivity (max degree = 4)
- ✓ Valid and DRC-compliant design



Recommendation: Fabricate the Linear topology first.

provides a cleaner, lower-congestion, and more robust starting point for fabrication, while the Star topology remains a strong candidate for experiments requiring central connectivity.